A Geometric Perspective on Polynomially Solvable Convex Maximization

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Agenda

1. Introduction
2. Comonotonicity
3. Polynomial-Time Solvability
4. Computational Experiments

Introduction

We consider

$$\max_{x \in \mathbb{R}^n} f(A^\top x) \quad \text{s.t. } x \in \mathcal{X} \tag{Conv Max}$$

where

- $f : \mathbb{R}^r \to \mathbb{R}$ is a finite convex function
- $A \in \mathbb{R}^{n \times r}$ has full column rank, where r is called the rank of the objective
- $\mathcal{X} \subseteq \mathbb{R}^n$ is the feasible region
- The optimal solution exists

Hardness

How hard is the problem?

$$\max_{x \in \mathbb{R}^n} f(A^\top x) \quad \text{s.t. } x \in \mathcal{X} \tag{Conv Max}$$

In general, NP-hard

- Maximizing a concave rank-2 quadratic function over a polytope is NP-hard
- There does not exist constant factor polynomial time approximation algorithm unless P=NP

Question: when is the convex maximization problem polynomially solvable?

One technical assumption

In this work, we work in fixed-rank settings

$$\max_{x \in \mathbb{R}^n} f(A^\top x) \quad \text{s.t. } x \in \mathcal{X} \tag{Conv Max}$$

For broad applicability, we assume that the problem is tractable once the support of x is fixed

Assumption (tractability under fixed support)

For any subset $S \subseteq [n]$, the subproblem

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \; & f(A^\top x) \\ \text{s.t. } \; & x \in \mathcal{X} \\ & \operatorname{supp}(x) = S \end{aligned}$$

is polynomially solvable, where $\operatorname{supp}(x) = \{i \in [n] : \; x_i \neq 0\}$.

- If $\mathcal{X} \subseteq \{0,1\}^n$, then the assumption holds trivially.

One technical assumption

- Sparsity constrained optimization problems

$$\max_{x \in \mathbb{R}^n} \left\{ \|A^\top x\|_2^2 : \|x\|_2 \leq 1, \|x\|_0 \leq k \right\} \qquad \text{(Single-SPCA)}$$

$\Rightarrow$

$$\lambda_{\max}(A_S A_S^\top) = \max_{x \in \mathbb{R}^n} \|A_S^\top x_S\|_2^2 \quad \text{s.t. } \|x_S\|_2 \leq 1 \qquad \text{(Subproblem)}$$

When the support is given, the problem reduces to ordinary PCA.

One technical assumption

- Manifold identification

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \ & \|A^\top x\|_2^2 \\ \text{s.t.} \ & \|x\|_2 = 1 \\ & x \geq 0. \end{aligned} \qquad \text{(Nonnegative-PCA)}$$

One technical assumption

- Manifold identification

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$\Rightarrow$

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \ & \|A^\top x\|_2^2 \\ \text{s.t.} \ & \|x\|_2 = 1 \\ & x_i > 0 \quad \forall i \in S \\ & x_i = 0 \quad \forall i \notin S \end{aligned} \qquad \text{(Subproblem)}$$

The feasible set is not closed due to the strict inequalities, but the subproblem can be handled by solving a small number of LPs.

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Comonotone Sets: A Motivating Example

- Let $\mathcal{X} = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$
- Consider the linear optimization problem

$$\max_{x \in \mathcal{X}} v^\top x$$

If $v_1 \geq v_2$ (or $v_1 \leq v_2$), then the above problem always admits an optimal solution x^* satisfying

$$x_1^* \geq x_2^* \text{ (or } x_1^* \leq x_2^*\text{)}.$$

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Observation: An interesting property

The order of the optimal solution x^* is determined by the order of cost vector v.

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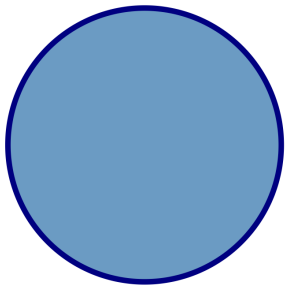
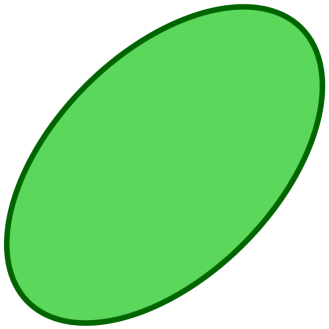
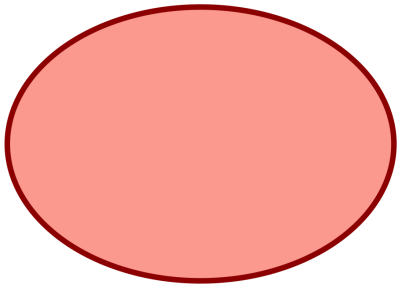
$x_1^2 + x_2^2 \leq 1$



$x_1^2 + x_2^2 - x_1 x_2 \leq 1$



$\frac{x_1^2}{2} + x_2^2 \leq 1$



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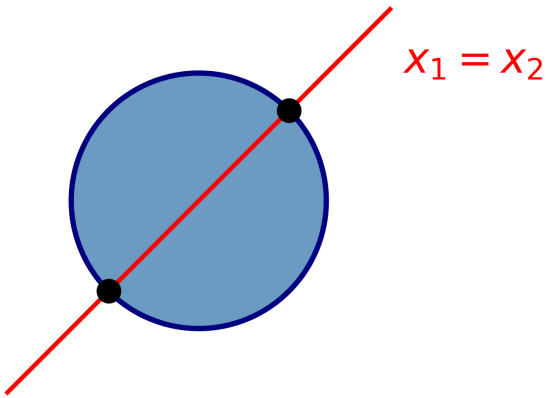

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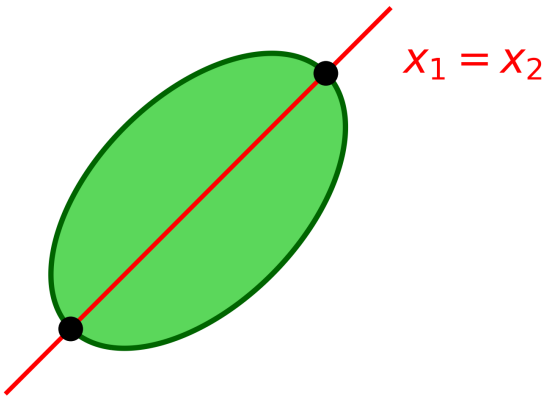

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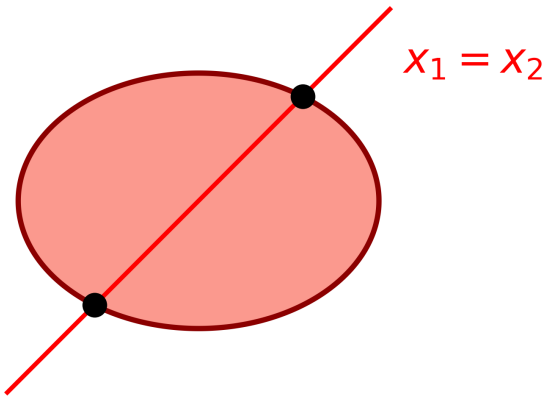

No

(If $v_1 \geq v_2$, the ordering of x^* varies)

Comonotone Sets: Definition

Definition

A set $\mathcal{X}$ is called comonotone if there is a mapping $T : \Pi_n \to \Pi_n$ such that given a permutation $\pi \in \Pi_n$, for any v satisfying

$$v_{\pi(1)} \geq \cdots \geq v_{\pi(n)},$$

the problem $\max_{x \in \mathcal{X}} v^\top x$ has an optimal solution x^* satisfying

$$x^*_{\sigma(1)} \geq \cdots \geq x^*_{\sigma(n)}, \quad \sigma = T(\pi).$$

If $T = I$ the identity mapping, then $\mathcal{X}$ is called standard comonotone.

Remark: $\mathcal{X}$ is standard comonotone if and only if

$$(v_i - v_j)(x^*_i - x^*_j) \geq 0 \; \forall i, j \in [n]$$

Standard Comonotone Sets

Question: what restriction does comonotonicity impose on $\mathcal{X}$?

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Theorem

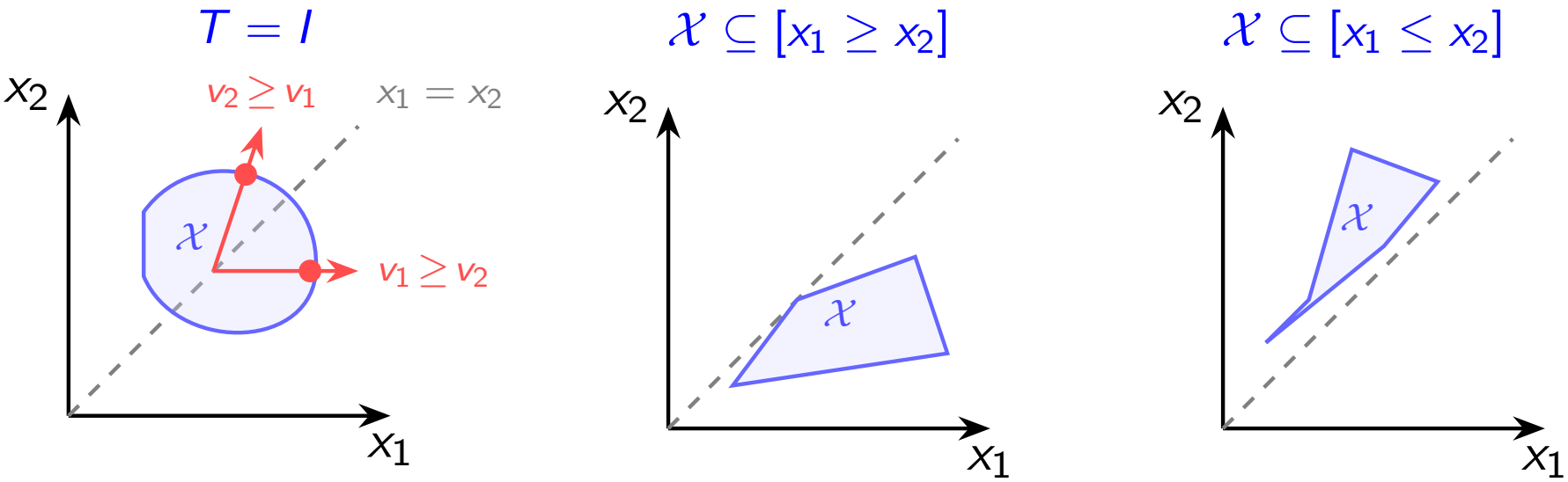
If $\mathcal{X}$ is comonotone and T is surjective, then $\mathcal{X}$ is standard comonotone.

Remark: Comonotonicity is a combinatorial rigidity

Standard Comonotone Sets

Implication in 2D space. If $\mathcal{X} \subseteq \mathbb{R}^2$ is comonotone, then

- $T = I$
- $T(\pi) = (1, 2)\ \forall \pi \in \Pi_2 \Leftrightarrow x_1 \geq x_2$ for all $x \in \mathcal{X}$
- $T(\pi) = (2, 1)\ \forall \pi \in \Pi_2 \Leftrightarrow x_1 \leq x_2$ for all $x \in \mathcal{X}$
- $T(1, 2) = (2, 1)$ and $T(2, 1) = (1, 2)$



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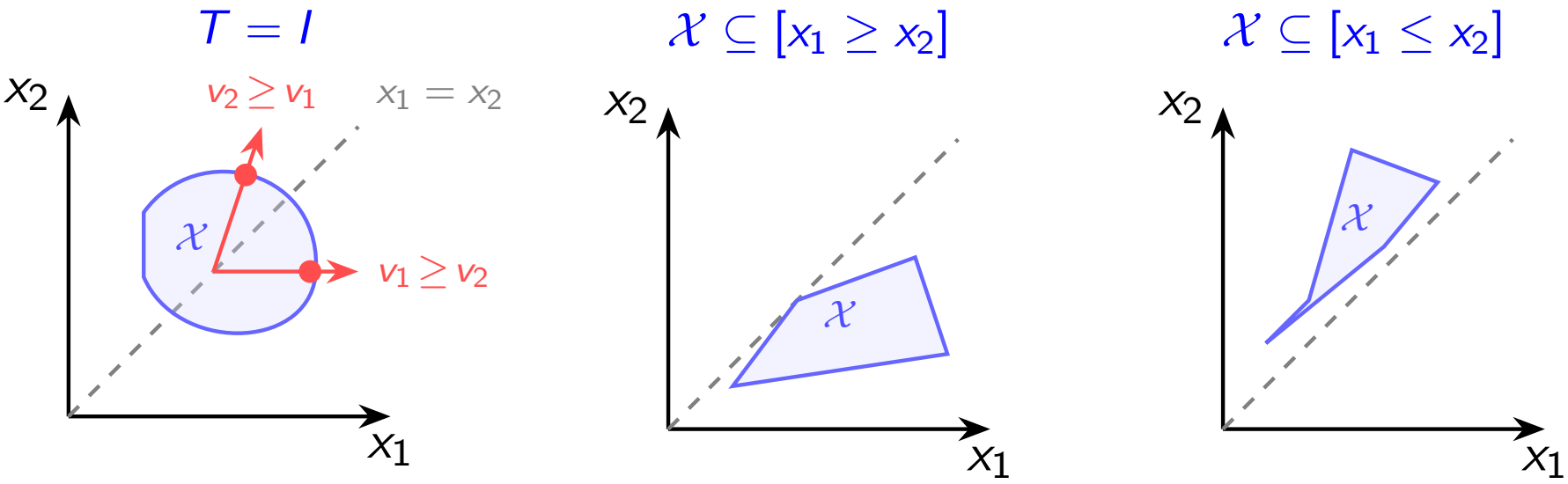
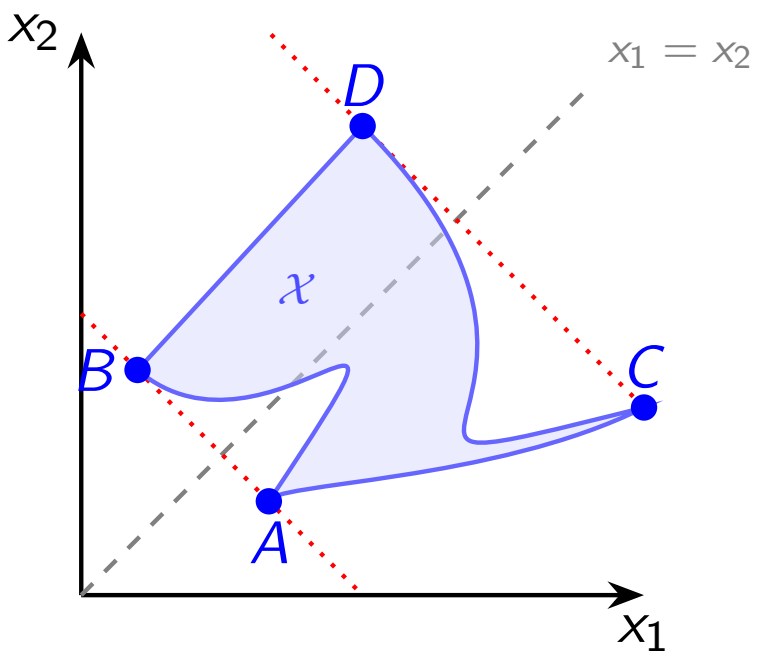


Standard Comonotone Sets

Implication in 2D space

Proposition

$\mathcal{X} \subseteq \mathbb{R}^2$ is standard comonotone iff for each $v \in \{(1,1),\ (-1,-1)\}$, the problem $\max\{v^\top x :\ x \in \mathcal{X}\}$ admits an optimal solution $\hat{x}$ with $\hat{x}_1 \geq \hat{x}_2$ and an optimal solution $\tilde{x}$ with $\tilde{x}_1 \leq \tilde{x}_2$.



Remark

- The ordering information is encoded by two "symmetric" directions.

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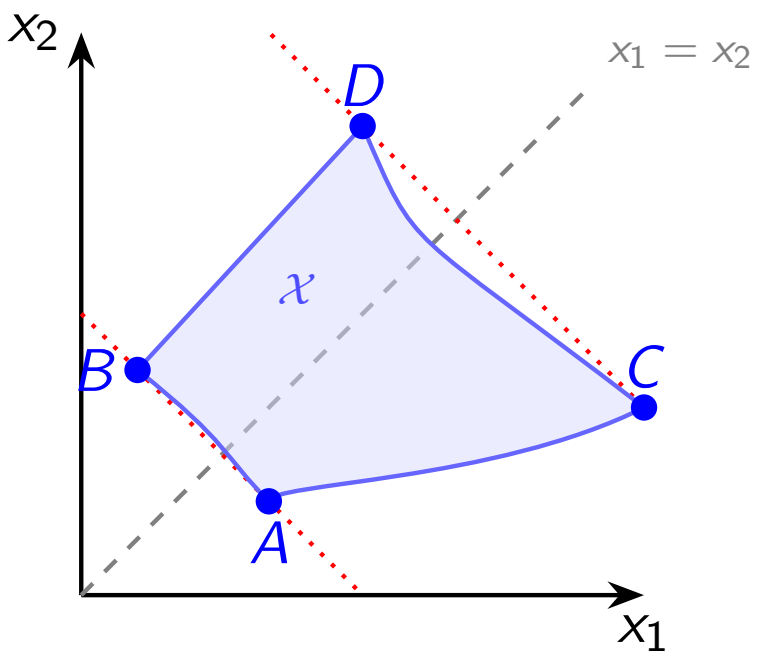

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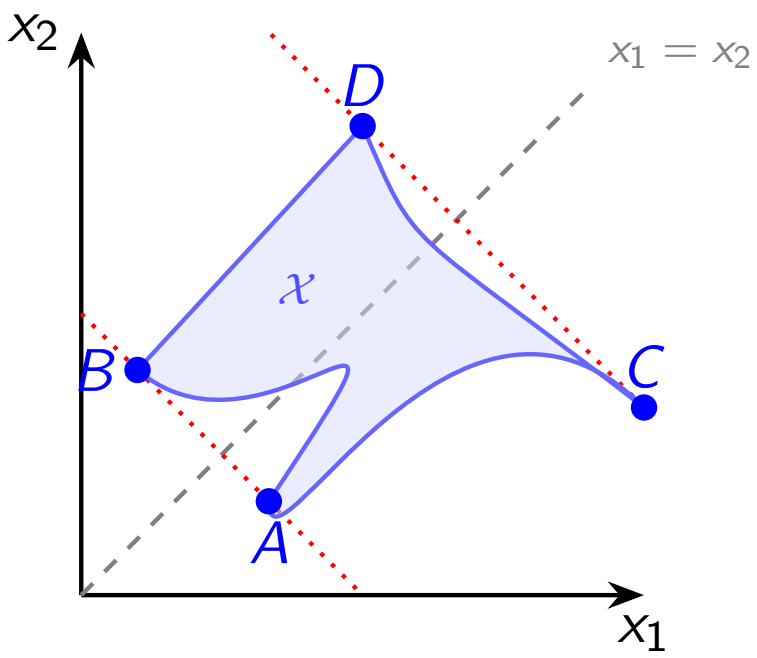

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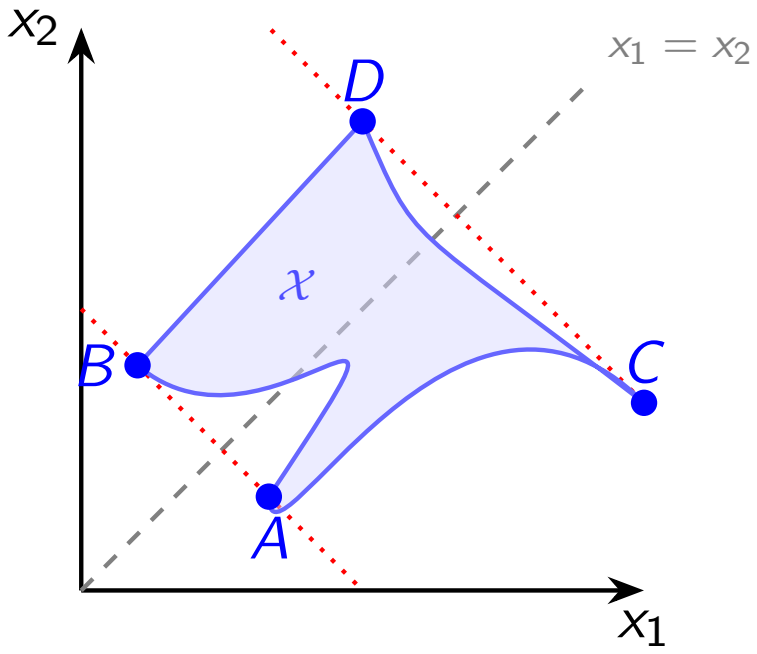

Remark

- To test standard comonotonicity, it suffices to solve two linear optimization problems.

Standard Comonotone Sets

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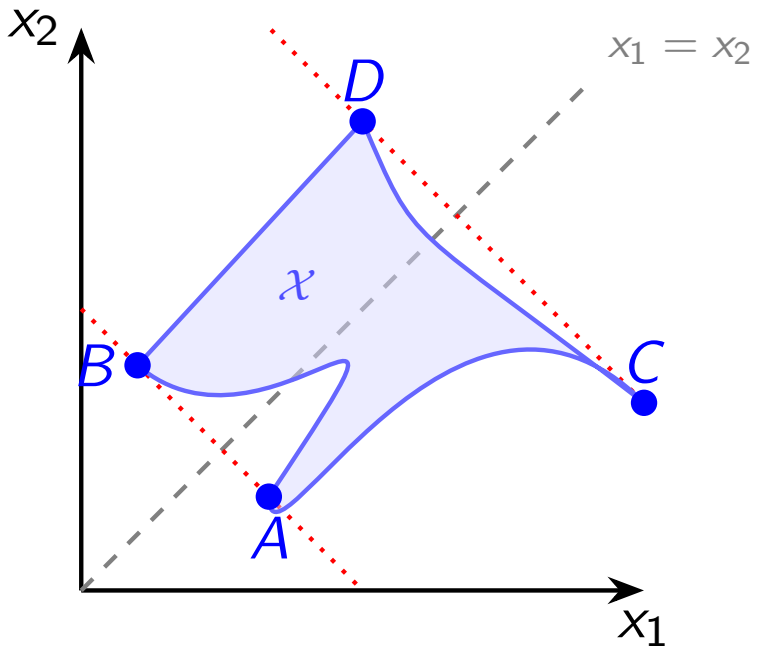


Remark

- The conclusion does not extend to higher dimensions in a straightforward way.

Standard Comonotone Sets

In general, one has the following characterization of standard comonotonicity.

Theorem

A set $\mathcal{X}$ is standard comonotone if and only if for all $v \in \mathbb{R}^n$ and $i, j \in [n]$ satisfying

$$v_i = v_j,$$

there exists an optimal solution $\bar{x}, \tilde{x} \in \arg\max\{v^\top x : \ x \in \mathcal{X}\}$ such that

$$\bar{x}_i \geq \bar{x}_j, \quad \tilde{x}_i \leq \tilde{x}_j.$$

Remark The symmetry on the dual side is mirrored on the primal side.

Standard Comonotone Sets

Remark

- A set $\mathcal{X} \subseteq \mathbb{R}^n$ is called permutation invariant if for every vector $x \in \mathcal{X}$ and $\pi \in \Pi_n$, the permuted vector $[x_{\pi(1)}, \ldots, x_{\pi(n)}] \in \mathcal{X}$. For example,

$$\mathcal{X} = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1,\ \|x\|_0 \leq k\}$$

Corollary

A permutation invariant set is standard comonotone.

- Standard comonotonicity is preserved under convexification.

Proposition

A set $\mathcal{X} \subseteq \mathbb{R}^n$ is standard comonotone if and only if its convex hull $\mathrm{conv}(\mathcal{X})$ *is standard comonotone.*

Standard Comonotone Sets

For convex sets, one has

Theorem

A convex set $\mathcal{X}$ is standard comonotone if and only if for all $v \in \mathbb{R}^n$ and $i, j \subseteq [n]$ satisfying

$$v_j = v_j,$$

there exists an optimal solution $x^* \in \arg\max\{v^\top x : \ x \in \mathcal{X}\}$ such that

$$x_i^* = x_j^*.$$

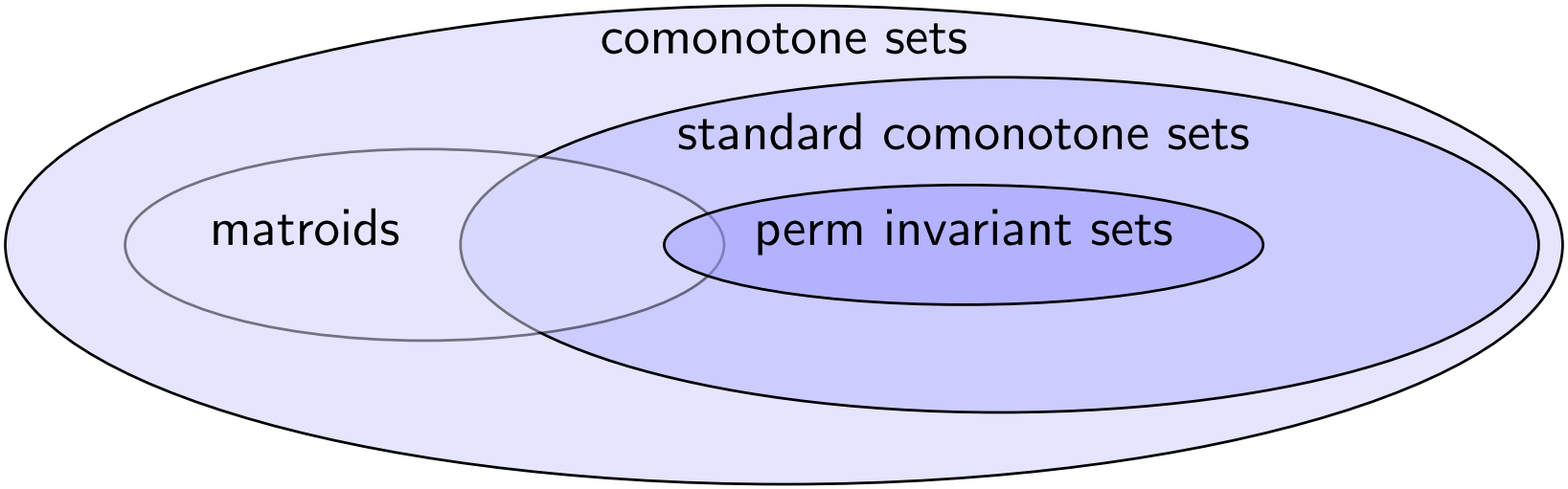
$\mathcal{X}$ is standard comonotone $\Leftrightarrow [v_i \geq v_j \Rightarrow x_i^* \geq x_j^*] \Leftrightarrow [v_i = v_j \Rightarrow x_i^* = x_j^*]$

Remark (conjugate symmetry): Ordering symmetry in the dual space is mirrored as a symmetry on the solution side in the primal space.

Non-standard comonotone sets

General comonotone sets can capture combinatorial structures that lack full symmetry. One can show the following sets are comonotone.

- Matroid: linearly independent sets, spanning trees
- Polymatroid
- M-convex sets
- Other greedy-related families ...



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Reduction to the Linear Counterpart

Back to the convex maximization problem

$$\max_{x \in \mathbb{R}^n} \left\{ f(A^\top x) : x \in \mathcal{X} \right\} \tag{P}$$

Proposition

There exists a vector $c \in \mathbb{R}^r$ such that every optimal solution to

$$\max_{x \in \mathbb{R}^n} \left\{ (Ac)^\top x : x \in \mathcal{X} \right\} \tag{LP}$$

is also optimal to (P).

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Issue: c is unknown $\Rightarrow$ aim to find the support of the optimal solution

Candidates supports

Consider

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- By the comonotonicity of $\mathcal{X}$,

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- For each fixed ordering of x^*

$$x^*_{\sigma(1)} \geq \cdots > x^*_{\sigma(t_1)} = \cdots = 0 = \cdots = x^*_{\sigma(t_2)} > \cdots \geq x^*_{\sigma(n)},$$

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Orderings of Ac

Note

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Orderings of Ac

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- For a general $v \in \mathbb{R}^n$, $\#[\text{orderings of } v] = n!$
- For $v \in \text{span}(A)$ where $A \in \mathbb{R}^{n \times r}$, not every ordering is possible

Example: $r = 1$

- $A = a \in \mathbb{R}^n$
- $v = Ac = ca$, where $c \in \mathbb{R}$
- Only two orderings of v, NOT $n!$
  - Same as a if $c \geq 0$
  - Opposite of a if $c < 0$

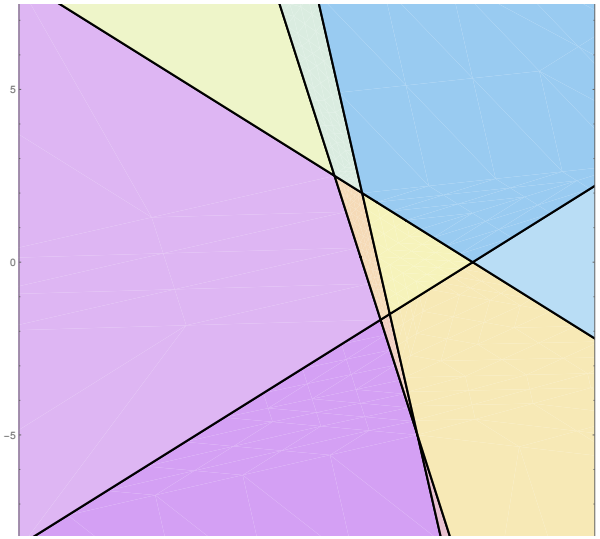
Hyperplane Arrangement

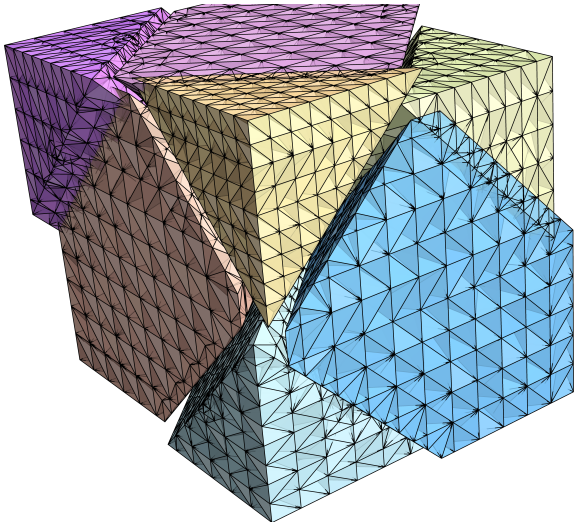
A collection of hyperplanes $\mathcal{H} = \{H_i^{=}\}_{i \in [p]}$

$$H_i^{=} \stackrel{\text{def}}{=} \left\{ u \in \mathbb{R}^q : \ (a_i)^\top u - b_i = 0 \right\},$$

the space $\mathbb{R}^q$ is partitioned into a family of cells.

- Each cell takes the form $\cap_{i \in [p]} H_i$, where $H_i \in \{H_i^{=}, H_i^{>}, H_i^{<}\}$.
- The collection of cells is called a hyperplane arrangement $\mathcal{A}(\mathcal{H})$.





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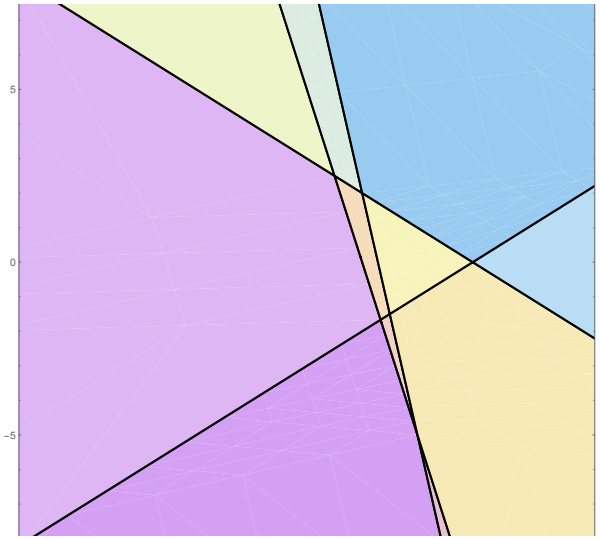
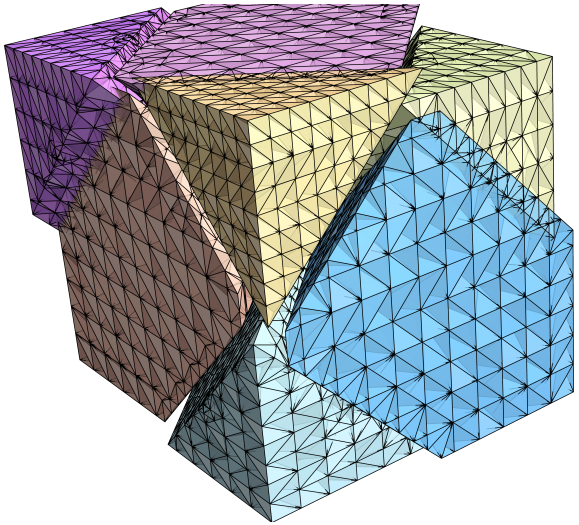
- Fact: $|\mathcal{A}(\mathcal{H})| = \mathcal{O}(p^q)$





Complexity for general comonotone sets

To apply, we consider the hyperplane arrangement induced by

$$\{c \in \mathbb{R}^r : \ (Ac)_i = (Ac)_j\} \quad \forall i \neq j \in [n].$$

- each hyperplane corresponds to one pairwise comparison
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Theorem

If $\mathcal{X}$ is comonotone, then the convex maximization problem

$$\max_x \{f(A^\top x) : \ x \in \mathcal{X}\}$$

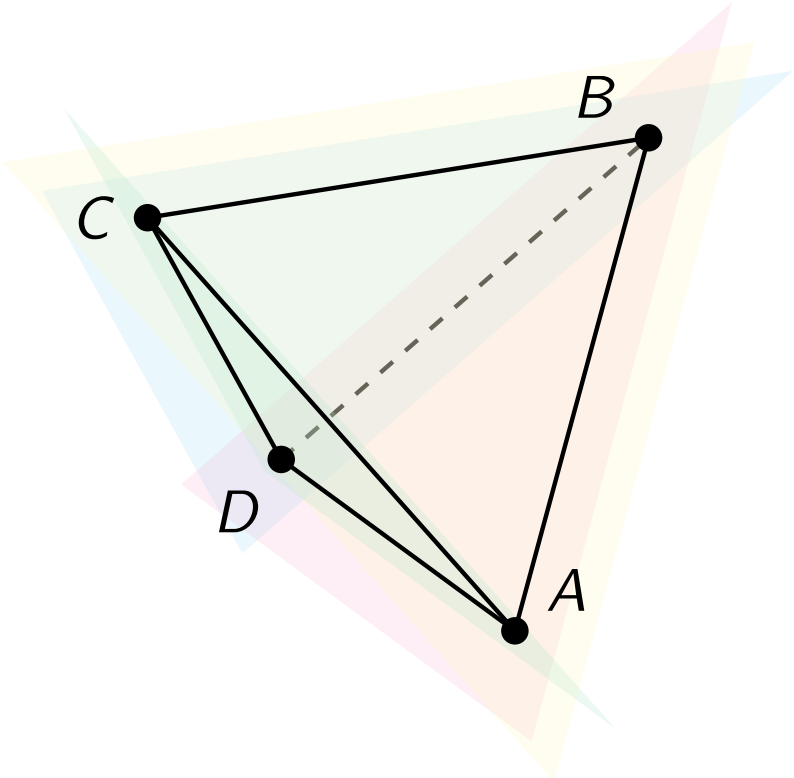
can be solved in $\mathcal{O}(n^{2r})$ oracle time.

A lifting technique under standard comonotonicity

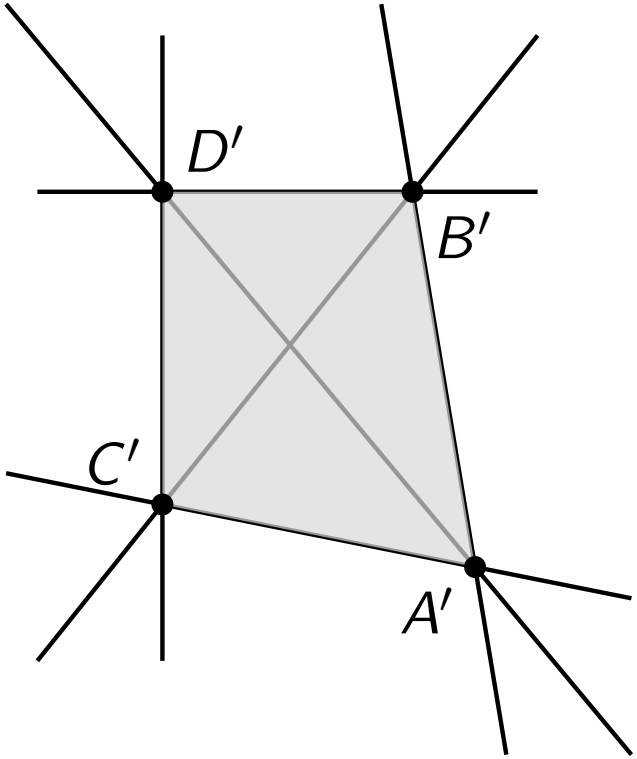
For standard comonotone sets, we consider the hyperplane arrangement

$$\{(c,t) \in \mathbb{R}^{r+1} : \ (Ac_i) = t\} \quad \forall i \in [n].$$

Lifted hyperplane arrangement



Original hyperplane arrangement



- The original HA is the projection of the lifted HA.

Complexity for standard comonotone sets

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More Comments

- The complexity can be further improved when more structures are available, e.g., when $\mathcal{X}$ is sign invariant
- The results can be readily extended to quasi convex objectives
- The results can be extended to "almost comonotone" sets

Table: Oracle complexity for application examples

	Single SPCA	Row SPCA	Disjoint SPCA	NN SPCA	TST
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More importantly, our method is much more implementable

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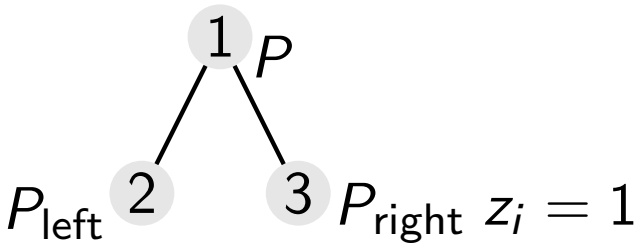

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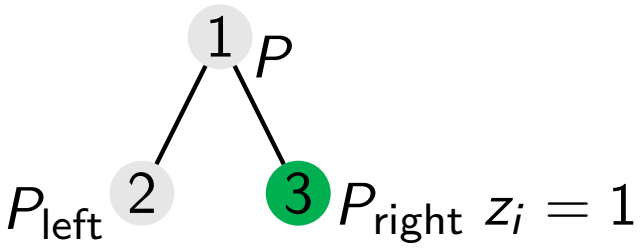

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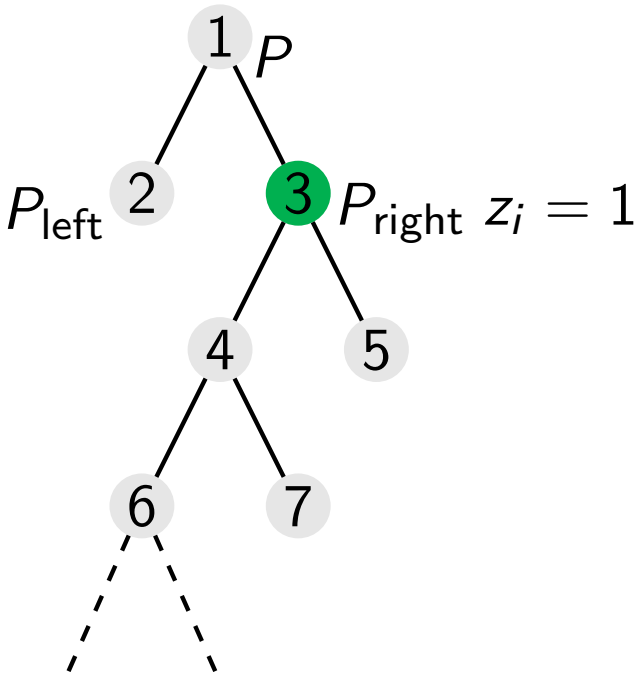

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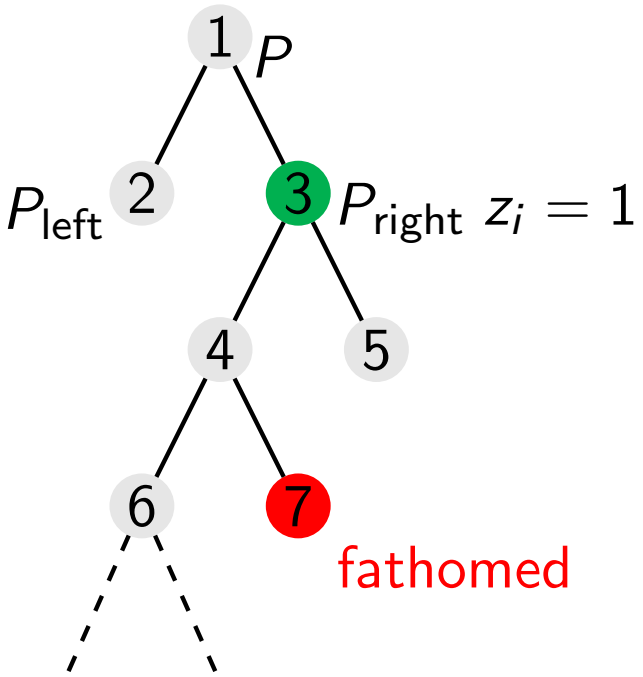

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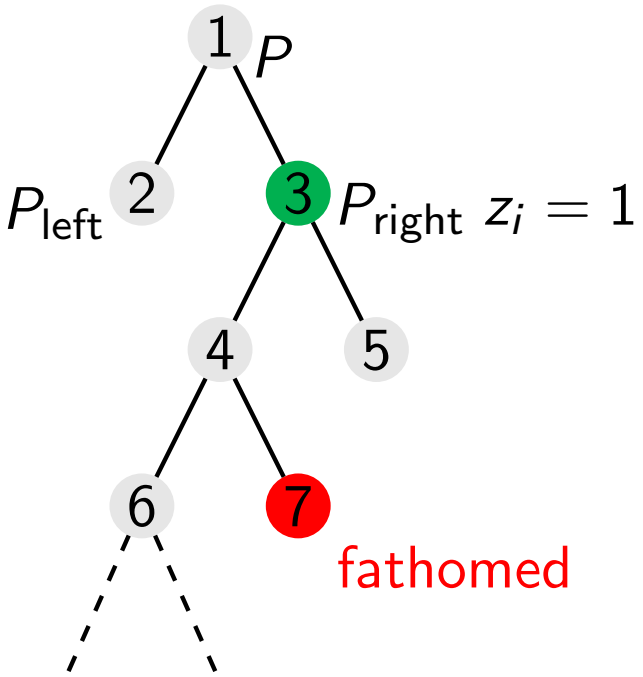

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- Each node requires solving one LP to test if $P = \emptyset$.
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- Parallel implementation? Memory explosion
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- The tree is very imbalanced
  - Restart BFS + DFS
- Very similar to Branch & Bound trees. All integer programming techniques can be applied to HA trees.

Agenda

1. Introduction
2. Comonotonicity
3. Polynomial-Time Solvability
4. Computational Experiments

Setup

We consider

$$\begin{aligned} \max\ & x^\top Q x \\ \text{s.t.}\ & \|x\|_2 = 1 \\ & \|x\|_0 \le k. \end{aligned}$$

To get a rank-r relaxation, assume $\lambda_1 \ge \cdots \ge \lambda_n$ and

$$Q = \lambda_1 u_1 u_1^\top + \cdots + \lambda_r u_r u_r^\top + \lambda_{r+1} u_{r+1} u_{r+1}^\top + \cdots + \lambda_n u_n u_n^\top.$$

We approximate Q with $\tilde{Q} \succeq Q$

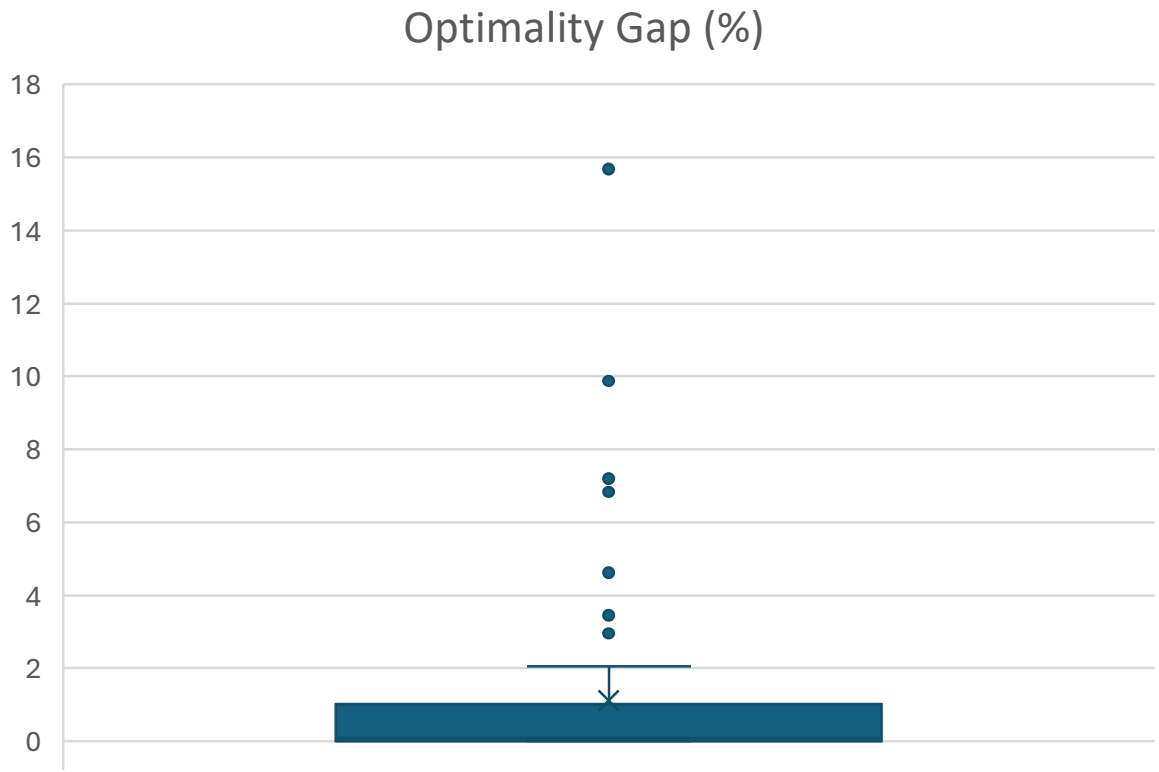
$$\tilde{Q} = \lambda_1 u_1 u_1^\top + \cdots + \lambda_r u_r u_r^\top + \lambda_{r+1} u_{r+1} u_{r+1}^\top + \cdots + \lambda_{r+1} u_{r+1} u_n^\top.$$

Observe

$$x^\top \tilde{Q} x = x^\top (\underbrace{\tilde{Q} - \lambda_{r+1} I}_{\text{rank}=r}) x + \lambda_{r+1} \ge x^\top Q x$$

Computational results - optimality gap

- $n \in [13, 128]$, $r \in [4, 13]$
- average gap $= 1.11\%$



Computational results - solution time

- Time limit: 3600 seconds
- Solvability: HA 97.9% versus Gurobi 76.6%

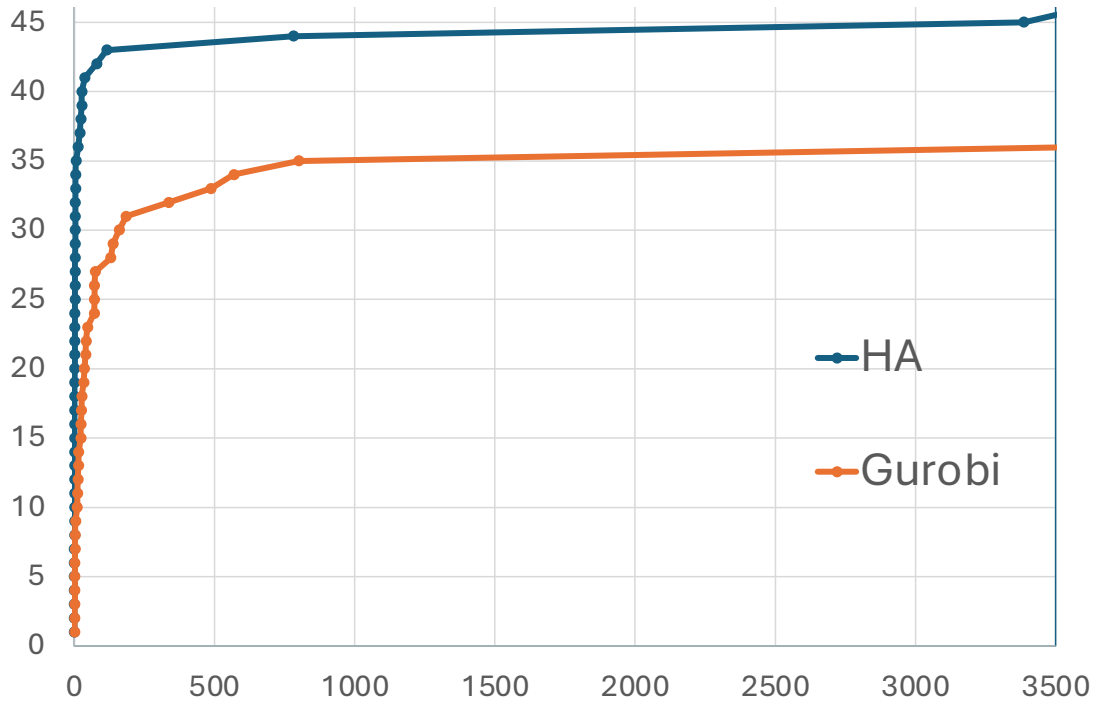


Figure: Computational Profile

Take Home Message

Summary

- Introduce comonotonicity – a type of symmetry defined through conjugate relationship
- Fixed-rank convex maximization problem over comonotone sets are polynomial-time solvable
- The proposed method achieves promising practical performance

Our paper is available at:

Thanks for your listening!